Temperature logger Bio3 Freiburg

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Charging

Plug to USB power source and set switch to ON to position.

Yellow light ON indicates charging.

Yellow light OFF indicates no charging.

Charging from empty to full takes 8 hours (400 mA current for 2600 mAh)

For charging, you can remove the SD-card, if you do not want to log data.

If you leave the SD-card in the circuit during charging, data logging is continuously active.

Switch on signals

Three blinks in green indicate battery is ok to go for at least a few days.

Three blinks in red indicate that the battery is close to empty.

The numbers of subsequent green blinks indicate the number of sensors recognized.

Afterwards, the first data frame is recorded and the logger is active. All LEDs are dark now.

If an older log data file exists, the new data will be appended. No data get ever deleted.

A permanent green light indicates NO SD card is recognized and the program

is aborted. In this case, switch the logger off, insert a good micro-SD card, and try again.

Use FAT16 for cards up to 2 GB and FAT32 for cards from 4 GB to 32 GB. The exFAT format is not

supported. Do not leave the logger with the fault-LED on, this will discharge the battery.

After the experiment, wash cable-bound sensors with demineralized water to remove salt and other contaminations. This is to prevent corrosion and to increase durability.

Change logging interval

Copy settings.txt to the SD card and edit it. Attention: This file is plain text and requires UNIX line ends (LF). Without this file present, the default logging interval is 300 seconds.

Temperature sensor accuracy of TMP117A

±0.1 ° C (maximum) from -20 ° C to 50 ° C

NIST traceability

Medical grade: meets ASTM E1112 and ISO 80601-2-56